Recognizing 3D Object Spoofing in Virtual Reality: Confidence and Repeated Exposure

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Abstract: Immersive virtual reality (VR) allows users to perceive 3D objects as part of a coherent spatial environment. However, this coherence may also make it difficult for users to recognize when virtual objects have been visually altered. We investigate the recognizability of altered 3D objects in VR, focusing on 3D object spoofing as a security-relevant form of alteration. Rather than measuring behavioral harm in realistic tasks, we evaluate an optimistic upper-bound setting in which users are explicitly asked to inspect objects and judge whether they are altered. We conducted a controlled VR study with 42 participants. Each participant completed 105 randomized trials across 21 conditions, including non-spoofed objects, 3D Wrapping, planar and mesh-based Shielding, different target shapes, and object placements. Recognition performance varied substantially across spoofing conditions. Fully covering Wrapping was particularly difficult to recognize and frequently produced false negatives. Confidence was generally associated with recognition performance, but for fully covering Wrapping, 90.48% of responses with high confidence were incorrect, and 72.62% were false negatives. Repeated exposure improved recognition performance for non-spoofed conditions but did not produce a corresponding improvement for Wrapping or Shielding. These findings indicate that human recognition alone may be insufficient for detecting 3D object spoofing and motivate countermeasures that incorporate system-side detection or verification.

1. Introduction

Virtual reality (VR) technology has rapidly advanced and is increasingly adopted in enterprise settings, including training and operational support in education, aerospace, and healthcare. By enabling immersion and spatial presence, VR can deliver realistic experiences that substantially improve task performance and user experience. At the same time, malicious manipulation in VR can have consequences that are more severe than in conventional computing environments such as PCs and smartphones, because users' perceptions and actions are tightly coupled to the virtual space and can translate into real-world decisions and physical behavior.

Spoofing is a particularly concerning threat in this context. In web and mobile contexts, phishing and impersonation are widely recognized as serious problems. In VR, an attacker can exploit immersion and spatial presence to influence user behavior, potentially escalating spoofing into attacks that involve physical safety risks. Fujita et al. demonstrated concrete forms of spoofing in VR and their technical feasibility[1], [2].

However, technical feasibility does not equate to perceivability. It remains dangerously unclear whether human users, immersed in a fabricated reality, can actually recognize these spoofed objects from authentic ones.

To obtain a conservative characterization of real-world risk, we frame our study as an optimistic upper bound on human detectability within our evaluated scope. Specifically, participants are explicitly prompted to look for spoofing, can actively inspect objects by moving their head, and encounter repeated conditions that allow attention focusing and strategy refinement , factors that tend to increase detection compared to realistic VR use where attention is divided by a primary task. If users fail to detect spoofing even under this "best-case" setting, relying on human perception alone is unlikely to be sufficient in practice.

This paper addresses this gap by empirically examining user recognizability of 3D object spoofing in VR. We focus on spoofing defined as a state in which an object different from the object that should be displayed is shown to the user. Following the definition and classification established by Fujita et al., we consider two primary forms of 3D object spoofing within Superimposition: Wrapping, which covers a target object in a three-dimensional manner, and Shielding, which blocks the user's view in a planar manner. We evaluate recognition performance across these spoofing forms and presentation conditions, and examine false negatives and form confusion as recognition errors.

We organize our investigation around the following research questions:

RQ1 Recognition Performance Across Spoofing Conditions: Under an optimistic upper-bound setting

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for human attention, how does recognition performance vary across spoofing forms and presentation conditions?

RQ2 Confidence and Recognition Performance: How is participants' confidence associated with recognition performance, and under which spoofing conditions do incorrect responses occur even when participants report high confidence?

RQ3 Repeated Exposure and Recognition Performance: Does repeated exposure to the same spoofing conditions improve recognition performance?

To answer these questions, we conduct a user study that quantitatively evaluates how likely users are to recognize different 3D object spoofing conditions. Beyond overall recognition rates, we analyze participants' response patterns to characterize false negatives and form confusion. We further examine the relationship between confidence and recognition performance and whether recognition performance changes across repeated presentations of the same conditions.

This paper makes three contributions. First, we provide an empirical evaluation of user recognition of Wrapping and Shielding in VR under an optimistic upper-bound setting, showing substantial differences in recognition performance across spoofing conditions and frequent false negatives under fully covering Wrapping. Second, we examine participants' confidence in their recognition judgments and show that incorrect responses can occur even with high confidence. In particular, fully covering Wrapping produced frequent high-confidence false negatives. Third, we examine recognition performance across repeated presentations and show that repeated exposure did not improve recognition performance for Wrapping or Shielding, although performance improved for non-spoofed conditions.

By clarifying conditions under which users struggle to recognize 3D object spoofing, and by examining confidence and repeated exposure together with recognition performance, this study provides a user-centered characterization of spoofing risks in VR.

2. Related Work

2.1 VR-Specific Security and Privacy Risks

VR introduces security and privacy risks that are specific to immersive, sensor-rich, and interactive 3D environments, including observables and manipulation surfaces beyond those in conventional computing environments.

Prior work has demonstrated VR-specific side channels. For example, Gopal et al. inferred user input by observing finger movements when users entered credentials on a virtual keypad [3], and Yang et al. inferred user input from avatar motions in a virtual space [4]. These findings suggest that signals tied to embodied interaction can leak sensitive user actions.

Another line of work studies perceptual manipulation attacks (PMA) [5], which exploit VR-specific perceptual characteristics and can steer users' behavior and thinking, potentially leading to physical harm and privacy violations [6], [7]. While these empirical studies primarily focus on visual manipulation, prior work also discusses the feasibility of PMA based on auditory stimuli [8]. Together, these studies motivate understanding not only the feasibility of VR attacks but also how they manifest to users under immersion and spatial presence.

2.2 Spoofing Risks in VR/AR

Spoofing has long been an important security concern in web, smartphone, and networked environments. Outside VR/AR, previous studies have examined spoofing-related attacks such as overlay attacks on Android [9] and defenses that apply ideas from active noise cancellation to visual noise [10]. In VR/AR, the high level of immersion and spatial presence in a 3D space can make the impact of spoofing more severe [2].

Fujita et al. organized the risk landscape of spoofing in VR and provided a taxonomy of 3D object spoofing that includes Superimposition-based attacks such as Wrapping (3D coverage of a target object) and Shielding (planar occlusion of a target object) [1], [2]. Importantly, their work discusses technical feasibility at runtime, where a malicious asset can manipulate object presentation after deployment and can generate, move, or deform objects while observing the scene state. However, this feasibility perspective does not directly answer whether users can recognize spoofed objects during immersive interaction.

3. Threat Model

We assume an adversary who can introduce or manipulate 3D content within an otherwise trusted VR application but cannot modify the user's system or data directly. Two deployment paths make this realistic. First, third-party assets distributed through marketplaces such as the Unity Asset Store[11], [12] or Fab [13] can embed modules that generate or transform objects at runtime; comparable marketplace-injection attacks have been reported for IDE and browser extensions[14], [15] , and sideloading via package managers can bypass store review. Second, in multi-user social VR such as VRChat [16], user-generated content can place or alter shared objects through legitimate authoring mechanisms. In both paths the adversary operates at low privilege and does not need to exploit software vulnerabilities.

Within this model, our experiments evaluate the runtime capability that both paths grant, namely superimposing a spoofing object onto a target. We instantiate the two superimposition forms defined by Fujita et al. [1], [2], Wrapping and Shielding, and vary the attacker-tunable parameters available at runtime: target geometry, placement of the spoofing object along the user-to-target axis, and parallax-aware scaling. We exclude higher-privilege vectors such as non-public API abuse and device rooting because they are not required for the manipulations we study.

4. Methodology

4.1 Study Objectives

Previous studies by Fujita et al. [2] organized the clas-

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sification and risks of spoofing and demonstrated the feasibility of spoofing against primitive objects. However, they did not evaluate user impact. In this study, we aim to clarify whether users can recognize spoofing in practice and to characterize the impact when spoofing occurs. Following Fujita et al.'s taxonomy and feasibility discussion, we evaluate recognizability for Wrapping and Shielding under an upper-bound quiz setting, thereby linking attack classification to task design.

To examine the extent to which users recognize spoofing, we conducted a user study that asked participants to perform spoofing recognition tasks.

In this study, we intentionally adopt a quiz-based recognition task to obtain an optimistic upper bound on human detectability of VR 3D object spoofing under our scope. Participants are explicitly instructed that spoofing may be present, are asked to judge spoofing on every trial, and can actively inspect the object by moving their head. Moreover, each condition is repeated multiple times, providing opportunities for attention, strategy refinement, and learning that would typically be absent in real deployments where users are engaged in a primary task and rarely question object authenticity. Therefore, if spoofing remains difficult to recognize in this "best-case" setting, it is unlikely to be more detectable in realistic scenarios with divided attention and higher cognitive load. We use this upper-bound framing to interpret our results as conservative with respect to real-world risk, while keeping the scope limited to the attack forms and stimuli evaluated in our study.

4.1.1 RQ1: Recognition Performance Across Spoofing Conditions

This objective corresponds to RQ1 by examining how recognition performance varies across spoofing forms and presentation conditions. We focus on false negatives, in which spoofed objects are judged as "Not spoofed," because these cases correspond to spoofing being accepted as legitimate. We also examine form confusion to characterize how parameters such as parallax-aware scaling, placement, and target geometry affect participants' responses.

Previous studies by Fujita et al. showed that spoofing can be implemented in different forms and with different parameters, but the extent to which users can recognize these variations remains unclear. We therefore compare response patterns across Wrapping and Shielding conditions to identify conditions that produce lower recognition performance or different types of recognition errors.

4.1.2 RQ2: Confidence and Recognition Performance

This objective corresponds to RQ2 by examining whether participants' confidence reflects their actual recognition performance. Because participants report confidence after every response, we analyze how accuracy changes across confidence levels and whether incorrect responses occur even when participants report high confidence.

In particular, we examine false negatives under high confidence because these responses indicate cases in which a spoofed object is judged as "Not spoofed" without substantial uncertainty. By comparing confidence across spoofing forms and conditions, we characterize conditions in which participants' subjective confidence and recognition performance differ.

4.1.3 RQ3: Repeated Exposure and Recognition Performance

This objective corresponds to RQ3 by examining whether repeated exposure to the same spoofing conditions changes recognition performance. Each quiz question is presented five times in randomized order, allowing us to compare responses from the first through fifth presentation of the same condition.

Because participants are explicitly informed about spoofing and repeatedly encounter the same conditions, this analysis provides an additional perspective on our upper-bound setting. We examine whether recognition performance improves across repeated presentations and whether such changes differ among Not spoofed, Wrapping, and Shielding conditions.

4.2 Study Overview

4.2.1 Task Design (Quiz-Based Spoofing Recognition)

To achieve the objectives described in the previous section, we designed a task in which participants answered quiz questions on spoofing recognition within a VR application developed by the authors (Fig. 1). In each question, participants viewed an object presented in front of them and selected whether it was spoofed and, if so, which spoofing form it represented. The answer options were: "Wrapping," "Shielding," "Spoofed but the type is unknown," "Not spoofed," and "Unknown." By distinguishing "Spoofed but the type is unknown" from "Unknown," we can separate errors attributable to difficulty in identifying the spoofing type from errors attributable to difficulty in recognizing spoofing presence. To measure response confidence, participants also rated their confidence on a five-point scale for each question. The application logged the correctness of each response, the selected option, and the confidence rating. HMD position and rotation and eye-tracking data were also recorded but are reserved for future analysis and are out of scope here.

Participants answered 21 unique quiz questions. To account for within-participant variability and to obtain more reliable estimates of recognition tendencies, we presented each question five times, resulting in 105 trials per participant. We fully randomized the trial order to avoid systematic ordering of the quiz questions. Full details of 21 question are provided in Appendix A.2. The design rationale for the 21 question patterns is described in the following section.

To reduce color-related confounds, we used a colorblind-friendly palette and ensured sufficient perceptual separability between stimulus colors. Full details of the color selection and discriminability checks are provided in Appendix A.3.

4.2.2 Post-Study Survey

After the VR task, participants completed a 39-item paper

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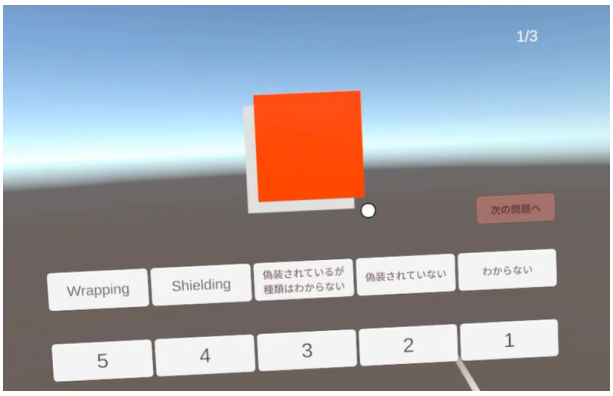


Fig. 1 Screenshot of the user study application running on Meta Quest Pro

survey covering perceived visual anomalies, trust and inspection strategies, VR experience, and relevant knowledge and experience. We use these responses as supplementary information for interpreting recognition performance and participants' reported inspection strategies. The full instrument is provided in the Appendix A.4.

4.3 Quiz Question Patterns and Conditions

The 21 quiz questions and their condition assignments are listed in the supplementary material. In the main text, we describe conditions at the factor level: (i) non-spoofed controls (4 shapes), (ii) Wrapping variants (Z-fighting, placement error, full coverage), (iii) plane-based Shielding with and without parallax-aware scaling, and (iv) mesh-based Shielding across target shapes and placements (1/4, 1/2, 3/4 of the user-to-target segment). Representative stimuli are provided in the Appendix A.5.

4.4 Procedure

The study was conducted by three authors. For each session, a single experimenter led the procedure. To ensure consistency across experimenters, we prepared a written script in advance.

Before the session began, the experimenter explained the study purpose and overview, the data we would collect, compensation, and the participant's right to stop the study at any time. Participants who agreed to take part then completed a consent form and a participant information form. The experimenter next explained the device and study details using a slide deck, including the definition and classification of spoofing and the overall study flow.

We then explained the eye-tracking setup procedure and how to answer the quiz questions. Participants practiced with tutorial questions to confirm the interaction flow. After the tutorial, we invited questions about the materials and procedure. We then handed the device to the participant to complete the eye-tracking setup. After calibration, we briefly collected the device to switch to the study application and configure the quiz question set. The participant then wore the device again and completed the in-app tutorial. After the tutorial, we again accepted questions, informed participants that the application would display their accuracy at the end of the main phase, and started the main phase. After participants completed the quiz, we collected the device and administered the post-study survey on paper. We also conducted a short oral debrief to collect feedback and impressions. Finally, participants completed a receipt form and received compensation.

4.5 Participants

We recruited participants both through social media and through on-campus recruitment, and we ran sessions with 42 participants. As an eligibility criterion, participants were required to have binocular visual acuity of at least 0.7, either uncorrected or corrected with glasses or contact lenses.

5. Results

5.1 Outcome definitions and analysis overview

User study quiz accuracy averaged 55.37%. Across participants, the highest accuracy was 91.43% (96/105) and the lowest was 29.52% (31/105).

To align with RQ1–RQ3, we analyze recognition performance from three perspectives. For RQ1, we examine correctness and response patterns across spoofing forms and conditions. We emphasize false negatives on spoofed trials, where spoofed objects are judged as "Not spoofed," because these cases correspond to spoofing being accepted as legitimate. We also examine form confusion, such as Shielding being judged as Wrapping, to characterize differences across spoofing parameters.

For RQ2, we examine the relationship between participants' confidence and recognition performance. Confidence was rated on a five-point scale for each response. In addition to examining accuracy across confidence levels, we treat confidence levels 4 and 5 as high-confidence responses and examine incorrect responses and false negatives under high confidence. We use a generalized estimating equation (GEE) model with response correctness as the dependent variable, confidence as an explanatory variable, quiz question ID as a covariate, and participant as the clustering variable.

For RQ3, we examine whether recognition performance changes across repeated presentations of the same question. Because each of the 21 questions was presented five times, we define the presentation number of each question from the first to fifth presentation for each participant. We compare accuracy across presentation numbers and use GEE models separately for each spoofing form, with response correctness as the dependent variable, presentation number as an explanatory variable, quiz question ID as a covariate, and participant as the clustering variable.

5.2 RQ1: Recognition performance across spoofing conditions

5.2.1 False negatives and overall recognition outcomes

We first examine recognition performance across spoofed and non-spoofed conditions.

Figure 2 shows, for each question, the proportion of re-

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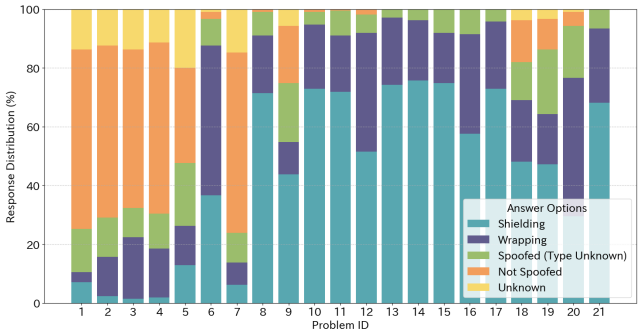


Fig. 2 Response option proportions by quiz question

sponses assigned to each answer option. For the Wrapping questions (Questions 5–7), participants frequently selected "Not spoofed." In particular, for Question 7, in which Wrapping fully covered the target object, "Not spoofed" accounted for 61.43% of the responses, whereas the correct response, "Wrapping," accounted for only 7.62%. This response pattern closely resembled those of the non-spoofed questions (Questions 1–4), indicating that participants often treated the fully covering Wrapping condition similarly to genuinely non-spoofed scenes.

In contrast, for the Shielding questions (Questions 8–21), "Shielding" was generally the most frequent response. However, responses also included "Wrapping" and "Spoofed but the type is unknown," indicating that participants did not always correctly identify the spoofing form even when spoofing was recognized.

Mean quiz accuracy was 57.86% for Not spoofed, 23.97% for Wrapping, and 61.39% for Shielding. Recognition accuracy differed significantly across quiz questions (Cochran's $Q(20) = 142.47, p < .001$). The lowest accuracy was observed for Question 7 (7.62%), followed by Question 5, which used Wrapping with Z-fighting (13.33%). These results indicate that Wrapping, especially the fully covering condition, produced substantially lower recognition performance and frequent false negatives.

5.2.2 Recognition differences by spoofing parameters

We next examine how response patterns differed across the parameters used in the spoofing conditions. For plane-based Shielding, Question 8 without parallax-aware scaling yielded an accuracy of 71.43%. In Question 9 with parallax-aware scaling, accuracy decreased to 43.81%. The proportion of "Not spoofed" responses increased from 0.95% to 19.52%, while "Spoofed but the type is unknown" increased from 8.10% to 20.00%.

Placement also affected recognition performance for mesh-based Shielding. Mean accuracy was 67.62% for the midpoint placement and 70.24% for the closer placement, whereas it decreased to 48.21% for the farther placement. Under the farther placement, responses were more frequently distributed across "Wrapping," "Spoofed but the type is unknown," and "Not spoofed," rather than being concentrated on the correct "Shielding" response.

Target geometry also affected response patterns. For mesh-based Shielding with a Cylinder, mean accuracy was 46.19%, and 40.48% of the responses were "Wrapping." For the other three target shapes, mean accuracy was 67.30%, and 20.85% of the responses were "Wrapping." Thus, the lower recognition performance for the Cylinder primarily reflected confusion between Shielding and Wrapping rather than an increase in false negatives.

Taken together, these results show that recognition performance varied across spoofing forms and presentation conditions. Wrapping produced frequent false negatives, whereas Shielding was more often recognized as spoofing but remained susceptible to form confusion depending on parallax-aware scaling, placement, and target geometry.

5.3 RQ2: Confidence and Recognition Performance

We next examine how participants' confidence was associated with recognition performance. Figure 3 shows recognition accuracy by confidence level and spoofing form. Accuracy generally increased with confidence. Mean accuracy was 22.76% for confidence level 1, 44.55% for level 2, 48.11% for level 3, 61.14% for level 4, and 63.45% for level 5.

To examine this relationship while accounting for differences across quiz questions and repeated responses from the same participants, we used a generalized estimating equation (GEE) model with a binary correct/incorrect response as the dependent variable, confidence as an explanatory variable, quiz question ID as a covariate, and participant as the clustering variable. Higher confidence was significantly associated with higher odds of a correct response (OR = 1.37, $p < .001$). These results indicate that confidence generally reflected recognition performance across the full set of questions.

However, high confidence did not necessarily indicate correct recognition under all spoofing conditions. When confidence levels 4 and 5 were grouped as high-confidence responses, the proportion of incorrect responses was 32.95% for the Not spoofed condition, 33.37% for Shielding, and 67.46% for Wrapping. Thus, more than half of the high-confidence responses under Wrapping were incorrect.

This tendency was particularly pronounced for Question 7, in which Wrapping fully covered the target object. Figure 4 shows the response distributions by confidence level for Question 7. As reported in RQ1, accuracy for this question was 7.62%, and 129 of 210 responses (61.43%) were "Not spoofed." Among the 84 responses with confidence levels 4 or 5, 76 (90.48%) were incorrect, and 61 (72.62%) were false negatives in which the spoofed object was judged as "Not spoofed." Even among responses with confidence level 5, 29 of 38 responses (76.32%) were "Not spoofed," while only 2 of 38 responses (5.26%) were correct.

Overall, these results indicate that participants' confidence was generally associated with recognition performance, but this relationship differed substantially across spoofing conditions. In particular, fully covering Wrapping frequently produced false negatives even when participants

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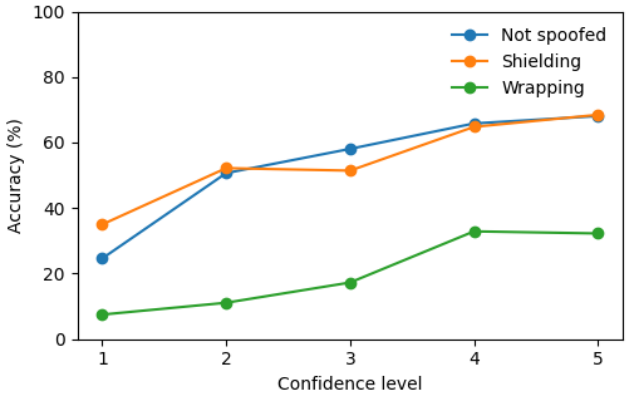


Fig. 3 Accuracy by confidence level and condition type

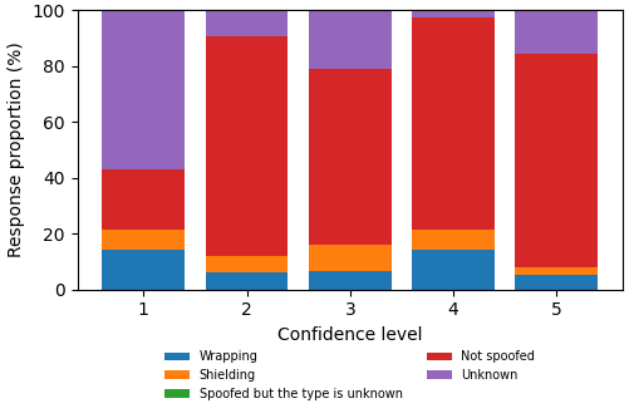


Fig. 4 Response option proportions by confidence level for Question 7 (fully covering Wrapping)

reported high confidence in their responses.

5.4 RQ3: Repeated Exposure and Recognition Performance

We next examine whether repeated exposure to the same spoofing conditions improved recognition performance. Each of the 21 quiz questions was presented five times. We therefore compared recognition accuracy from the first through fifth presentation of each question.

Overall accuracy increased from 50.68% on the first presentation to 56.12% on the second, 56.35% on the third, 56.80% on the fourth, and 56.92% on the fifth. However, the change in accuracy differed across spoofing forms.

Figure 5 shows accuracy across the five presentations separately for Not spoofed, Wrapping, and Shielding. For the Not spoofed questions, accuracy increased from 32.74% on the first presentation to 67.26% on the fifth presentation. In contrast, accuracy for Wrapping remained between 23.02% and 26.19% across the five presentations. Accuracy for Shielding also remained relatively stable, ranging from 60.37% to 62.59%.

To examine these changes while accounting for differences across quiz questions and repeated responses from the same participants, we used GEE models separately for each spoofing form, with a binary correct/incorrect response as the dependent variable, presentation number as an explanatory

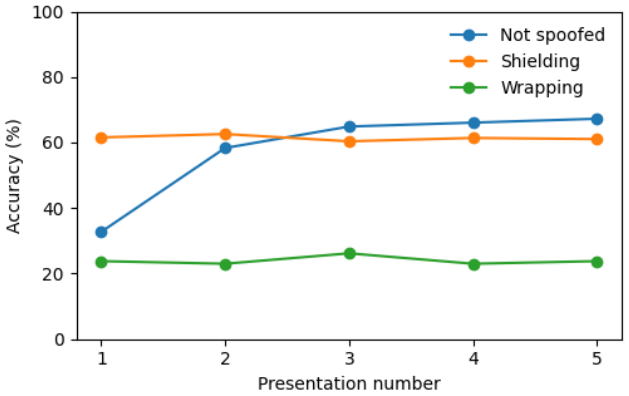


Fig. 5 Accuracy across repeated presentations by condition type

variable, quiz question ID as a covariate, and participant as the clustering variable. For the Not spoofed questions, later presentations were significantly associated with higher odds of a correct response (OR = 1.39 per presentation, $p < .001$). In contrast, no significant association was observed for Wrapping (OR = 1.00, $p = 1.000$) or Shielding (OR = 0.99, $p = .804$).

Taken together, these results indicate that the increase in overall accuracy across repeated presentations was primarily attributable to improved performance for the Not spoofed questions. Repeated exposure did not produce a corresponding improvement in recognition performance for either Wrapping or Shielding.

5.5 Summary of key findings

Across conditions, participants' recognition performance varied substantially. Wrapping, especially the fully covering case (Question 7), produced frequent false negatives, whereas Shielding was more often recognized as spoofing but remained susceptible to form confusion depending on parallax-aware scaling, placement, and target geometry.

Participants' confidence was generally associated with recognition performance, but high-confidence responses did not always indicate correct recognition. This tendency was particularly pronounced for Question 7, where 90.48% of responses with confidence levels 4 or 5 were incorrect, and 72.62% were false negatives. Repeated presentations also did not produce a corresponding improvement in recognition performance for Wrapping or Shielding, although accuracy increased for the Not spoofed questions.

Taken together, these results show that spoofing can remain difficult to recognize even when participants actively inspect objects, report high confidence in their judgments, and repeatedly encounter the same conditions.

6. Discussion

6.1 Recognition Difficulty Across Spoofing Conditions

The results show substantial differences in recognition performance across spoofing conditions. Wrapping was particularly difficult for participants to recognize, with a mean

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accuracy of 23.97%. In particular, Question 7, where Wrapping fully covered the target object, yielded an accuracy of 7.62%, the lowest among all questions. Moreover, participants frequently selected "Not spoofed," indicating that the fully covering Wrapping condition was often treated similarly to a genuinely non-spoofed object.

One possible explanation is that fully covering Wrapping can preserve geometric consistency around the target object. Because the target is completely covered, users may not observe depth discontinuities, occlusion boundaries, or motion-parallax inconsistencies even when actively inspecting the scene. This suggests that spoofing that preserves geometric coherence can remain difficult to recognize through human perception alone.

Shielding showed higher recognition performance than Wrapping, but response patterns varied depending on presentation conditions. Farther placement reduced accuracy and increased responses other than the correct "Shielding" response. Target geometry also affected form identification, particularly for Cylinder, where participants more frequently selected "Wrapping." These results indicate that Shielding is more likely to be recognized as spoofing, but its form can still be misidentified depending on spatial and geometric conditions.

6.2 Confidence and Recognition Performance

Confidence was generally associated with recognition performance across the full set of questions. However, this relationship differed substantially across spoofing conditions. Under Wrapping, incorrect responses remained frequent even when participants reported high confidence.

This pattern was most evident for fully covering Wrapping. For Question 7, 90.48% of responses with confidence levels 4 or 5 were incorrect, and 72.62% were false negatives. Thus, participants did not simply fail to recognize the spoofing while being uncertain about their judgments. In many trials, they reported high confidence while judging the spoofed object as "Not spoofed."

From a security perspective, this result indicates a limitation of relying on users to recognize suspicious visual states. If recognition failures were primarily accompanied by low confidence or "Unknown" responses, uncertainty itself could provide an indication that additional verification is needed. In contrast, the high-confidence false negatives observed under fully covering Wrapping show that some recognition failures may occur without such subjective uncertainty. Therefore, user confidence alone cannot be used as an indication that a displayed object has been correctly recognized.

6.3 Effect of Repeated Exposure

Participants encountered each quiz question five times in randomized order. This repetition provided multiple opportunities to inspect the same spoofing conditions and to adjust their recognition strategies. Overall accuracy increased across repeated presentations, but the increase was primarily observed for the Not spoofed questions.

In contrast, recognition performance for Wrapping and Shielding did not significantly improve across repeated presentations. This result is notable because participants were explicitly informed about spoofing and repeatedly encountered the same conditions within the study. Even under these favorable conditions, repeated exposure alone did not lead to improved recognition of either spoofing form.

These results further support the upper-bound interpretation of our study. In realistic VR use, users are unlikely to repeatedly inspect the same object while actively considering the possibility of spoofing. Our results therefore suggest that repeated exposure to spoofed objects should not be assumed to make such manipulation easier to recognize.

6.4 Implications for Countermeasures and Defense Strategies

Our results indicate that countermeasures against 3D object spoofing should not rely on human recognition alone. In particular, fully covering Wrapping produced frequent false negatives even under an upper-bound recognition setting, and some of these false negatives occurred with high confidence. Repeated exposure also did not improve recognition performance for Wrapping or Shielding.

One direction is therefore to prevent or detect suspicious object manipulation at the system or platform level. As discussed in previous studies, asset vetting and dynamic sandboxing can be used to identify suspicious behaviors in third-party content before distribution or execution. Runtime detection can also monitor abnormal object creation, deletion, transform changes, and rendering or occlusion behaviors that may indicate spoofing. For cases in which the expected visual state can be determined, integrity guarantees or verification of what should be rendered may provide stronger protection than relying on users to detect visual inconsistencies.

The results also suggest that simply making visual anomalies noticeable may not be sufficient. For example, Question 5 produced low recognition accuracy even though flicker was frequently reported as a noticeable visual anomaly in the post-study survey. Thus, noticing an unusual visual state does not necessarily result in correct recognition of spoofing. User-facing warnings or other visual cues should therefore be considered together with system-side detection or verification mechanisms rather than as a replacement for them.

7. Future Work

Future work should evaluate recognizability under more realistic targets, including complex meshes, transparency, and textured materials, and in task contexts where users' attention is divided. The evaluation should also move beyond recognition performance to examine whether undetected spoofing changes user behavior or leads to unsafe or unintended actions.

In addition, our repeated-presentation analysis was limited to repeated exposure within a single study session. Fu-

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ture work should examine whether longer-term experience or explicit training changes recognition performance.

8. Limitations

Our sample of 42 participants reveals clear condition-level trends but limits power for fine-grained subgroup analyses, and skews toward younger, technology-affine users, so external validity to broader populations remains open. Results are also specific to a Unity implementation on Meta Quest Pro and to primitive shapes with controlled materials; stereoscopic rendering, engine details, and the absence of complex meshes, textures, and transparency all shape the perceptual cues available, and replication across devices, engines, and richer assets is needed.

Methodologically, the quiz-based task is a deliberate optimistic upper bound: participants were informed about spoofing, could focus on inspection, and repeatedly encountered the same conditions. Realistic deployments with divided attention and higher cognitive load may therefore yield lower recognition performance. However, the repeated-presentation analysis captures repeated exposure only within a single study session and should not be interpreted as an evaluation of long-term learning or explicit training.

Finally, confidence was measured using participants' self-reported ratings after each response, and we measured recognizability rather than downstream behavior. We also did not isolate the causal perceptual mechanisms that produced recognition failures. Whether undetected spoofing changes user decisions, and which cues drive recognition under each condition, are left to future work.

9. Conclusion

We conducted a user study to examine the recognizability of 3D object spoofing in VR under an optimistic upper-bound setting. Recognition performance varied substantially across spoofing conditions. Fully covering Wrapping was particularly difficult to recognize and frequently produced false negatives, whereas Shielding was more often recognized as spoofing but remained susceptible to form confusion depending on presentation conditions.

We further found that high confidence did not always correspond to correct recognition, particularly under fully covering Wrapping, and that repeated exposure did not improve recognition performance for Wrapping or Shielding. These results indicate that human recognition alone may be insufficient for detecting 3D object spoofing and motivate countermeasures that incorporate system-side detection or verification.

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Appendix

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A.1 Ethical Considerations

We conducted the study based on informed consent. Participants were informed that they could withdraw from the study at any time without penalty. Before the session, we explained the study purpose, procedures, expected duration, and the types of data to be collected. Written consent was obtained from all participants prior to participation.

Participants received compensation in the form of a 2,000 JPY Amazon gift card.

The experimental tasks involved visual inspection and interaction within a VR environment. The study was carefully designed so that no private or personally sensitive information of participants would be recorded during the experiment.

After explaining the details of data collection, we obtained explicit written consent from participants before collecting interaction logs (e.g., quiz responses, response times, and headset pose data). Eye-tracking data, when recorded, were collected only after obtaining the same explicit consent. These data were stored securely and used only for research purposes.

A.2 Quiz questions

Table A·1 summarizes the 21 quiz questions, including the spoofing form, target shape, and condition assigned to each question.

A.3 Color Petterns

To improve visual accessibility and reduce bias related to color recognition, we adopted a standardized color scheme. We selected stimulus colors from the "Accent Color" set in the Color Universal Design (CUD) guidelines[17], which refines a colorblind-friendly palette originally proposed by Okabe and Ito (as summarized by Wong [18]). From the nine accent colors, we used seven: red, yellow, green, pink, orange, purple, and brown. We excluded sky blue and blue to maintain sufficient contrast with the experimental background.

We assessed the discriminability of the selected colors using the CIEDE2000 color difference metric. For every pair among the seven colors, the color difference (ΔE_{00}) was at least 3.0, a commonly used threshold for perceptible differences. This design reduces the likelihood that color discriminability becomes a confounding factor in the recognition task. To further avoid systematic artifacts, we randomized the assignment of colors to the target and spoofing objects across five material patterns for each of the 21 color pairs.

A.4 Post-study survey items

Tables A·2 and A·3 list the 39 items included in the post-study survey and their response formats.

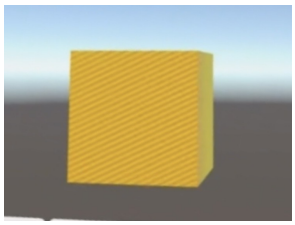


Fig. A·1 Question 5: Wrapping with Z-fighting

A.5 Quiz question patterns and conditions in detail

This section provides additional details on the 21 quiz questions and the corresponding spoofing conditions used in the user study.

Questions 1–4 present non-spoofed objects and serve as a control condition for quantitative analyses of recognizing spoofing presence. We informed participants during the tutorial that some questions would include non-spoofed objects.

Question 5 presents a Wrapping case with Z-fighting (Figure A·1). Z-fighting can occur when two objects are placed with nearly identical position, rotation, and scale, causing flickering due to ambiguity in rendering order. We included this case to examine whether users recognize spoofing under this implementation artifact.

Question 6 presents a Wrapping placement failure in which the target object is partially exposed. This intentionally unnatural case tests whether participants reliably recognize spoofing when the spoofing object is visibly misaligned.

Question 7 presents a Wrapping case that fully covers the target object (Figure A·2). Because the target cannot be seen from any viewpoint, this case tests whether participants mistakenly judge the scene as not spoofed.

Question 8 uses Plane-based Shielding without parallax-aware scaling (Figure A·3), allowing apparent downscaling due to stereo viewing. We included it to measure how this downscaling affects spoofing recognition.

Question 9 uses the same Plane-based Shielding but scales the plane based on stereo disparity so that the target object is fully occluded (Figure A·4). Comparing Questions 8 and 9 isolates the effect of parallax-aware scaling on recognition difficulty.

Questions 10–13 use mesh-based Shielding (Figures A·5–A·6) with method variants corresponding to each target primitive object. We placed the spoofing object at the midpoint between the user and the target (Figure A·7).

Questions 14–17 use the same mesh-based Shielding but place the spoofing object closer to the user (at 1/4 of the segment from the user to the target; Figures A·8–A·11). Questions 18–21 place it farther from the user (at 3/4 of the segment; Figures A·12–A·15). These placements allow us to examine how placement affects users' ability to recognize Shielding.

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Table A·1 Quiz questions presented in the experimental application

Question ID	Spoofing form	Target shape	Description
1	Not spoofed	Cube	
2	Not spoofed	Sphere	
3	Not spoofed	Cylinder	
4	Not spoofed	Capsule	
5	Wrapping	Cube	Wrapping with Z-fighting
6	Wrapping	Cube	Wrapping with a placement error of the spoofing object
7	Wrapping	Cube	Wrapping that fully covers the target object
8	Shielding	Cube	Plane-based Shielding without parallax-aware scaling for apparent downscaling
9	Shielding	Cube	Plane-based Shielding with parallax-aware scaling to compensate for apparent downscaling
10	Shielding	Cube	Mesh-based Shielding
11	Shielding	Sphere	Mesh-based Shielding
12	Shielding	Cylinder	Mesh-based Shielding
13	Shielding	Capsule	Mesh-based Shielding
14	Shielding	Cube	Mesh-based Shielding (closer placement than Question 10)
15	Shielding	Sphere	Mesh-based Shielding (closer placement than Question 11)
16	Shielding	Cylinder	Mesh-based Shielding (closer placement than Question 12)
17	Shielding	Capsule	Mesh-based Shielding (closer placement than Question 13)
18	Shielding	Cube	Mesh-based Shielding (farther placement than Question 10)
19	Shielding	Sphere	Mesh-based Shielding (farther placement than Question 11)
20	Shielding	Cylinder	Mesh-based Shielding (farther placement than Question 12)
21	Shielding	Capsule	Mesh-based Shielding (farther placement than Question 13)

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Table A·2 Post-study survey items and response formats

Item ID	Item text	Response format
1	The outline of the presented object sometimes appeared misaligned with the background or surrounding area.	7-point Likert
2	The boundary of the presented object sometimes appeared doubled.	7-point Likert
3	The rendering of object edges sometimes appeared jagged or insufficiently smooth.	7-point Likert
4	Circular or spherical objects sometimes appeared angular.	7-point Likert
5	Circular or spherical objects sometimes appeared angular.	7-point Likert
6	Another object sometimes appeared to protrude from the presented object.	7-point Likert
7	The surface of the presented object sometimes appeared to flicker or blink.	7-point Likert
8	The direction or intensity of shadows on the presented object sometimes did not match other objects.	7-point Likert
9	Highlights or reflections on the presented object sometimes appeared unnatural.	7-point Likert
10	Shading or perceived surface relief on the presented object sometimes appeared unnatural.	7-point Likert
11	The texture or resolution of the presented object sometimes differed from its surroundings and appeared unnatural.	7-point Likert
12	When I moved my head, the presented object sometimes behaved unnaturally relative to my viewpoint.	7-point Likert
13	I sometimes felt that it was difficult to focus or to bring the object into focus.	7-point Likert
14	The images seen by the left and right eyes (parallax) sometimes did not match, making depth appear unnatural.	7-point Likert
15	The visuals sometimes did not move smoothly and appeared choppy.	7-point Likert
16	The visual resolution sometimes appeared coarse, making the outline of the presented object look jagged.	7-point Likert
17	The material appearance or texture of the presented object sometimes did not match the surrounding environment.	7-point Likert
18	The presented objects and information felt natural.	7-point Likert
19	Even when the presented object looked natural, I tried not to trust it.	7-point Likert
20	I was able to stay focused during the study.	7-point Likert
21	I carefully inspected the outline of the presented object.	7-point Likert
22	I moved my head to observe how the presented object looked.	7-point Likert
23	I carefully inspected shadows and reflections of the presented object.	7-point Likert
24	If you noticed any other specific cues that felt unnatural, please describe them.	Free text
25	If you felt that you answered a question incorrectly, please describe why.	Free text
26	Did you feel that spoofing became easier to recognize as you answered more questions?	Free text
27	Which cues did you prioritize the most?	Free text
28	Approximately how many years ago did you first experience VR?	Free text
29	How frequently do you currently use VR?	Multiple choice (5 options)
30	When you use VR, what is your typical session duration?	Multiple choice (5 options)

Table A·3 Post-study survey items and response formats (continued)

Item ID	Item text	Response format
31	What are your primary purposes for using VR? (Select all that apply.)	Multiple choice (7 options)
32	Do you have experience with game development or 3D modeling?	5-point Likert
33	Do you have knowledge related to computer graphics (CG)?	5-point Likert
34	Do you have knowledge related to security?	5-point Likert
35	When observing things in everyday life, do you tend to focus on details?	5-point Likert
36	Have you ever experienced phishing harm (e.g., personal data leakage or financial loss)?	Multiple choice (3 options)
37	How confident are you in identifying phishing?	7-point Likert
38	Have you received education or training on phishing countermeasures? (Select all that apply.)	Multiple choice (3 options)
39	In the past year, have you ever suspected phishing and actually reported or responded to it?	Multiple choice (4 options)

A.6 Additional Results

This section provides additional analyses of recognition performance that supplement the results reported in Section 5.

We examined whether recognition accuracy differed by spoofing form using a generalized linear mixed model (GLMM) with a binary correct/incorrect response as the dependent variable, spoofing form as a fixed effect, and participant as a random effect (Figure A·17). We interpret odds ratios as relative odds of a correct response: values below 1 indicate lower odds of being correct (harder), whereas values above 1 indicate higher odds (easier), relative to the reference form ("Not spoofed").

Because Cochran's Q test indicated significant differences in accuracy across quiz questions, we further analyzed

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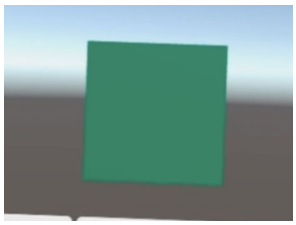


Fig. A·2 Question 7: Wrapping that fully covers the target object

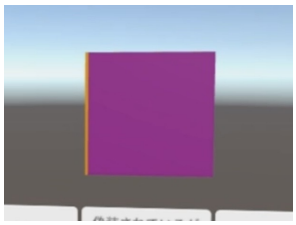


Fig. A·3 Question 8: Plane-based Shielding without parallax-aware scaling

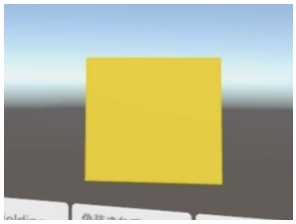


Fig. A·4 Question 9: Plane-based Shielding with parallax-aware scaling

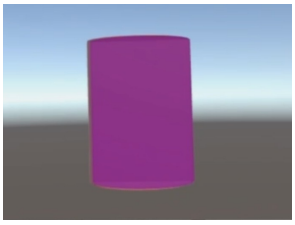


Fig. A·5 Question 12: Mesh-based Shielding for a Cylinder

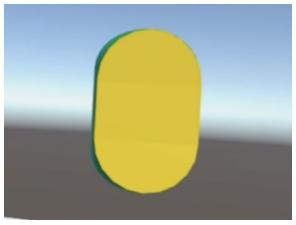


Fig. A·6 Question 13: Mesh-based Shielding for a Capsule

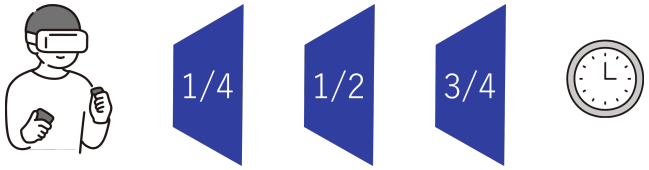


Fig. A·7 Placement of mesh-based Shielding objects. The left marker indicates the user, and the right marker indicates the target object. Let the user position be 0 and the target position be 1. In Questions 10–13, the spoofing object is placed at the midpoint (1/2). In Questions 14–17, it is placed at 1/4. In Questions 18–21, it is placed at 3/4.

question-level differences. Using the same model, we replaced the fixed effect with quiz question ID while keeping participant as a random effect, and we sorted questions by the estimated odds ratio (Figure A·18). Compared with accuracy by question ID (Figure A·16), the ordering was identical except that Questions 4 and 16 were swapped.

The highest estimated odds ratios were observed for Question 14 (mesh-based Shielding for a Cube, closer to the user),

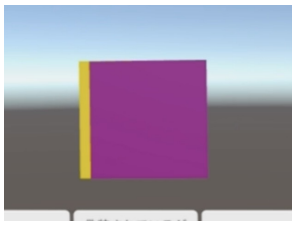


Fig. A·8 Question 14: Mesh-based Shielding for a Cube (closer to the user)

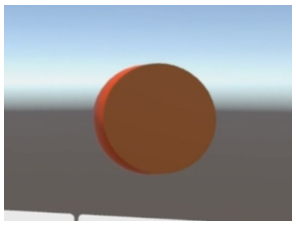


Fig. A·9 Question 15: Mesh-based Shielding for a Sphere (closer to the user)

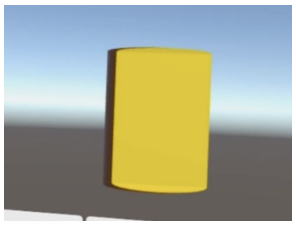


Fig. A·10 Question 16: Mesh-based Shielding for a Cylinder (closer to the user)

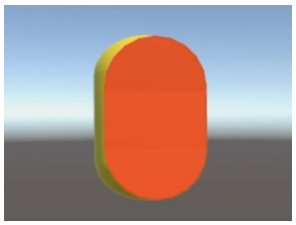


Fig. A·11 Question 17: Mesh-based Shielding for a Capsule (closer to the user)



Fig. A·12 Question 18: Mesh-based Shielding for a Cube (farther from the user)

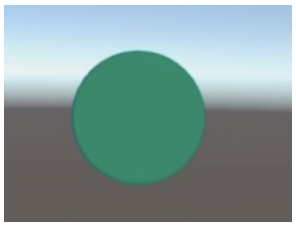


Fig. A·13 Question 19: Mesh-based Shielding for a Sphere (farther from the user)

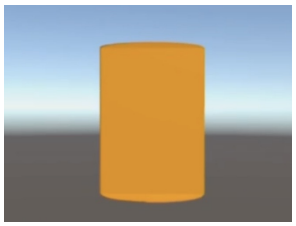


Fig. A·14 Question 20: Mesh-based Shielding for a Cylinder (farther from the user)

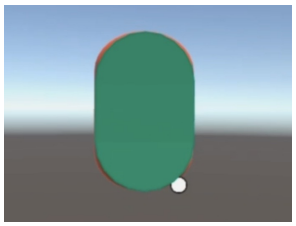


Fig. A·15 Question 21: Mesh-based Shielding for a Capsule (farther from the user)

Question 15 (mesh-based Shielding for a Sphere, closer to the user), and Question 13 (mesh-based Shielding for a Capsule). The lowest odds ratios were observed for Question 7 (Wrapping that fully covers the target object), Question 5 (Wrapping with Z-fighting), and Question 20 (mesh-based Shielding for a Cylinder, farther from the user).

We also estimated item difficulty parameters using item response theory (IRT) (Figure A·19). The highest estimated difficulty was for Question 7, followed by Questions 5 and 20. In contrast, questions using mesh-based Shielding were estimated to have relatively low difficulty. The GLMM and IRT results showed consistent patterns: the Wrapping questions (Questions 5, 6, and 7) had low odds of a correct response

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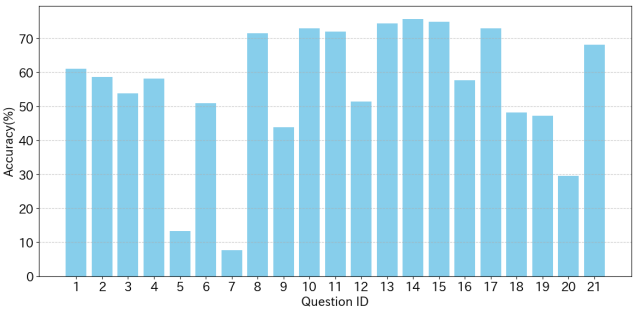


Fig. A·16 Accuracy by question ID

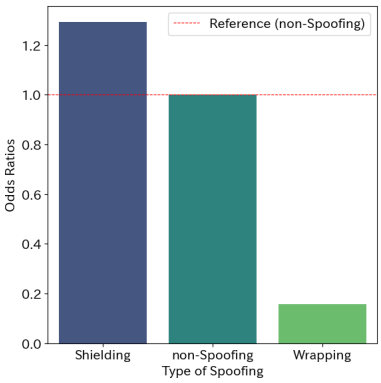


Fig. A·17 Odds ratios by spoofing form estimated by the GLMM

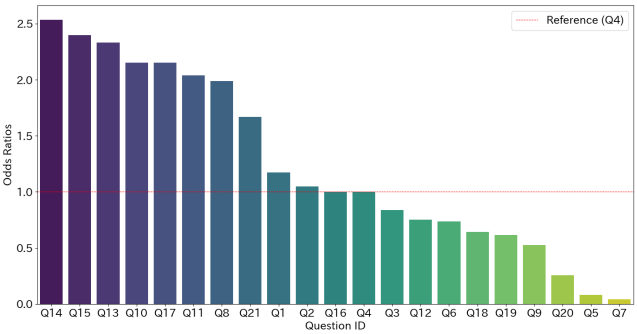


Fig. A·18 Odds ratios by quiz question estimated by the GLMM

and were also estimated as high-difficulty items under IRT.

Overall, these results indicate clear differences in recognition difficulty across quiz questions. In particular, Wrapping exhibited lower accuracy and lower odds of a correct response and was also estimated as higher difficulty under IRT. In contrast, mesh-based Shielding showed relatively higher accuracy and lower estimated difficulty. Importantly, the most difficult Wrapping conditions were also those in which spoofed objects were frequently judged as "Not spoofed" (i.e., false negatives), implying that spoofing can be accepted as legitimate even under our upper-bound recognition setting.

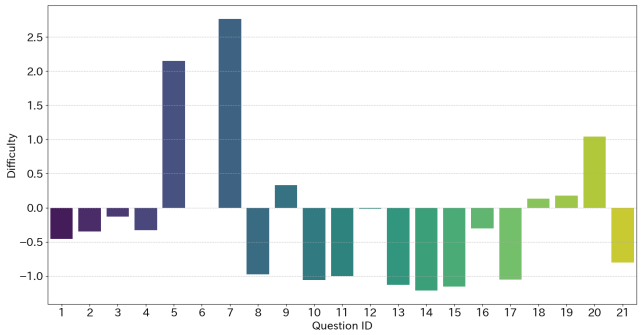


Fig. A·19 Estimated difficulty parameters from IRT (1PL)

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